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Lab 3: Component Modelling & Architectural Pattern Selection

Microservices Architecture — 2 Reasons

1. Independent Services:

The system can be divided into separate services such as User Management, Ride Management, Ride Matching, Expense Splitter, and Payment Service. Each service can work independently.

2. Easy Scalability:

Different services can be scaled independently. For example, if many users are searching for rides, the **Ride Matching** service can be scaled without scaling the entire system.

Security Advantage — 1 Reason

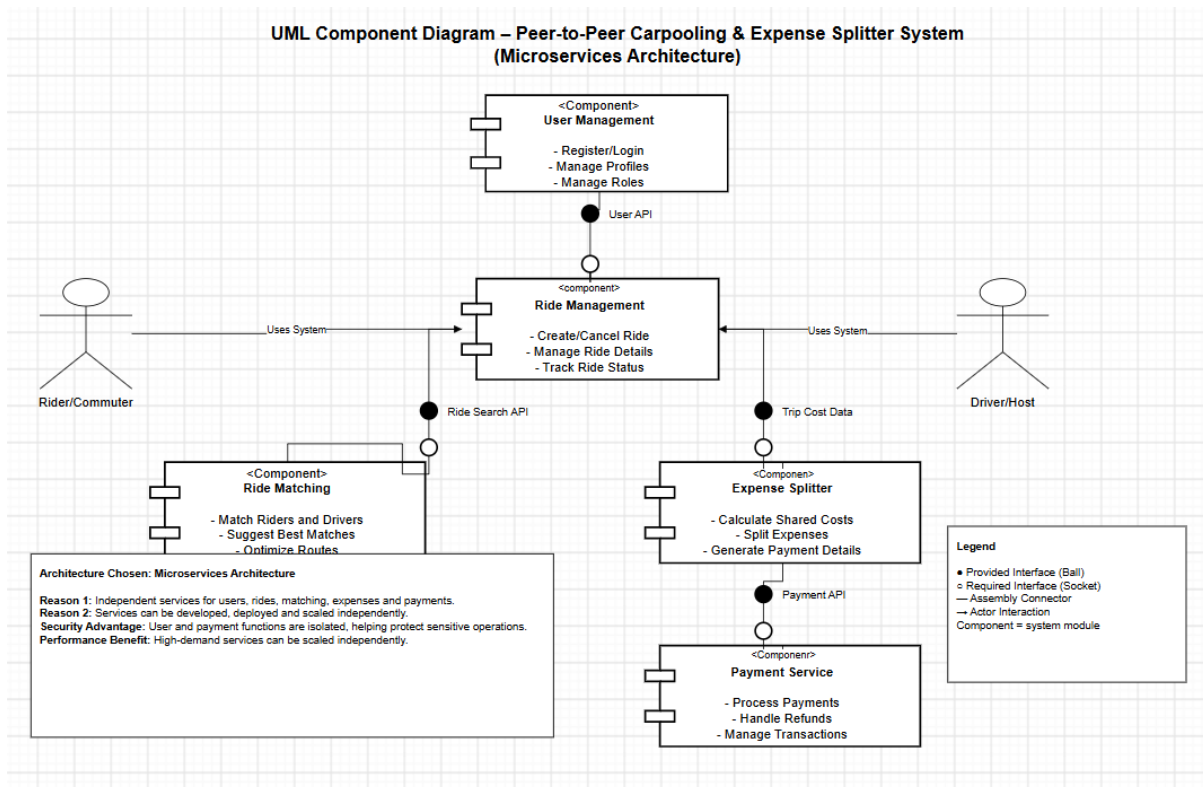
Service Isolation:

User and payment-related functions are separated into different services, which helps protect sensitive user and payment information and limits unauthorized access.

Performance Benefit — 1 Reason

Independent Processing:

Different services can process tasks separately, so heavily used services such as Ride Matching can handle more requests without slowing down the entire system.



Components Used:

No.	Component	Main Function
1	User Management	Register/login and manage Rider/Driver profiles
2	Ride Management	Create, cancel and manage rides
3	Ride Matching	Match riders with suitable drivers
4	Expense Splitter	Calculate and split trip expenses
5	Payment Service	Process the rider's payment

Interfaces Used:

No.	Interface	Provided By	Required By
1	User API	User Management	Ride Management
2	Ride Search API	Ride Management	Ride Matching
3	Trip Cost Data	Ride Management	Expense Splitter
4	Payment API	Expense Splitter	Payment Service

